



VECTOR SPACES

Deepthi Rao

Department of Science & Humanities

CLASS 9 : CONTENT



- Four Fundamental subspaces

FOUR FUNDAMENTAL SUBSPACES OF A MATRICES



Let A be the matrix of order $m \times n$. Associated with it are four subspaces which are defined as follows

1. $C(A)$ is the column space of A is a subspace of $\mathbb{R}^m$ and contains all the linear combination of the column vectors of A .

If then $\rho(A) = k$ then $\dim(C(A)) = k$

A basis of $C(A)$ corresponds to the columns having the pivots in echelon form of A .

2. $C(A^T)$ are the row space of A is a subspace of $\mathbb{R}^n$ and contains all the linear combinations of the rows of A .

If $\rho(A) = k$ then $\dim(C(A^T)) = k$

VECTOR SPACES

FOUR FUNDAMENTAL SUBSPACES OF A MATRICES



A basis for $C(A^T)$ is the set of row vectors in A or in the echelon form corresponding to the pivots in the echelon form.

3. $N(A)$ are the null space of A consists of all the solutions of the system $Ax = 0$. It is a subspace of $\mathbb{R}^n$.

If $\rho(A) = k$ then $\dim(N(A)) = n - k$

A basis for $N(A)$ is obtained by solving the system $Ux = 0$, identifying the pivot variables and free variables where U is the row reduced echelon form of A i. e. special solutions to $Ux = 0$ forms the basis of $N(A)$

4. $N(A^T)$ are the left null spaces of A is a subspace of $\mathbb{R}^n$ and consists of all the solution to the system $A^T x = 0$.

VECTOR SPACES

FOUR FUNDAMENTAL SUBSPACES OF A MATRICES



If $\rho(A) = k$ then $\dim(N(A^T)) = m - k$.

A basis for $N(A^T)$ is obtained by looking at the zero rows of U and then tracing back to the corresponding rows of A .

VECTOR SPACES

FOUR FUNDAMENTAL SUBSPACES OF A MATRICES



Note :

1. The row space of $A_{m \times n}$ is the column space of A^T .
It is spanned by the rows of A .
2. The left null space contains all vectors y for which $A^T y = 0$.
3. $N(A)$ and $C(A^T)$ are subspaces of R^m
4. $N(A^T)$ and $C(A)$ are subspaces of R^n
5. $\text{Dim } C(A) = \text{Dim } C(A^T) = r = \text{rank of } A$
6. $\text{Dim } N(A) = n - r$ and $\text{Dim } N(A^T) = m - r$.
7. The dimension of the null space of a matrix is called its nullity.

VECTOR SPACES

FOUR FUNDAMENTAL SUBSPACES OF A MATRICES



The rank- nullity theorem :

For any matrix $A_{m \times n}$,

$\dim C(A) + \dim N(A) = \text{no. of columns}$ i.e

$$r + (n-r) = n$$

This law applies to as well.

Hence, $\dim C(A^T) + \dim N(A) = m$ i. e

$$r + (m-r) = m$$

VECTOR SPACES

FOUR FUNDAMENTAL SUBSPACES OF A MATRICES



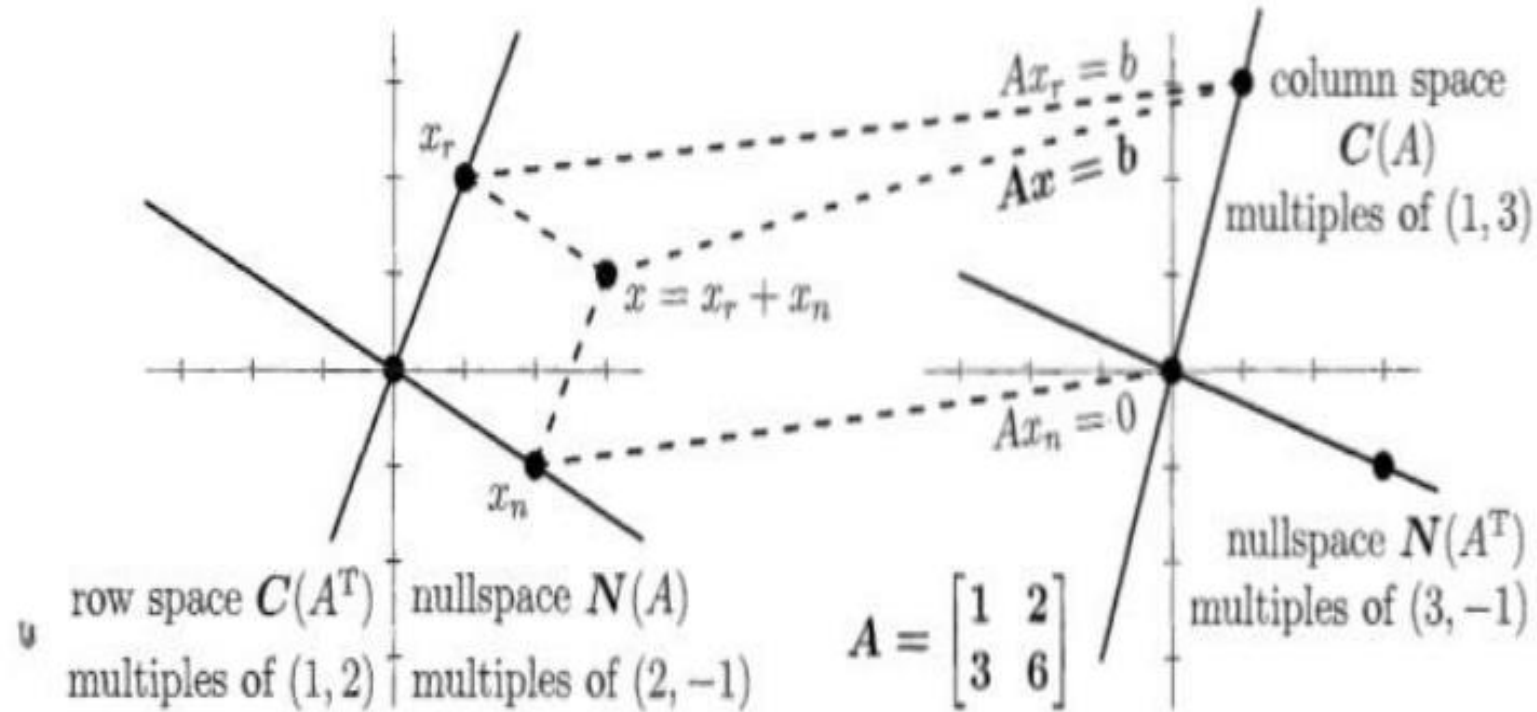
Let $A = \begin{bmatrix} 1 & 2 \\ 3 & 6 \end{bmatrix}$

Then, $m = n = 2$ and $\text{rank } r = 1$.

1. $C(A)$ is the line through $(1, 3)$
2. $C(A^T)$ is the line through $(1, 2)$
3. $N(A)$ is the line through $(-2, 1)$
4. $N(A^T)$ is the line through $(-3, 1)$

VECTOR SPACES

FOUR FUNDAMENTAL SUBSPACES OF A MATRICES



VECTOR SPACES

FOUR FUNDAMENTAL SUBSPACES OF A MATRICES



E.g. : Find the dimensions and a basis each for the four fundamental subspaces of the matrix.

Solution

$$A = \begin{bmatrix} 1 & 2 & 1 & 2 \\ 1 & 2 & 1 & 3 \\ 3 & 6 & 3 & 7 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & 1 & 2 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\sim \begin{bmatrix} 1 & 2 & 1 & 2 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}; C(A) = \left\{ c_1 \begin{bmatrix} 1 \\ 1 \\ 3 \end{bmatrix} + c_2 \begin{bmatrix} 2 \\ 3 \\ 7 \end{bmatrix} / c_1, c_2 \in R \right\}$$

VECTOR SPACES

FOUR FUNDAMENTAL SUBSPACES OF A MATRICES



$$\text{Basis for } C(A) = \left\{ \begin{bmatrix} 1 \\ 1 \\ 3 \end{bmatrix}, \begin{bmatrix} 2 \\ 3 \\ 7 \end{bmatrix} \right\}; \dim C(A) = \text{Rank of } A$$

$$c(A^T) = \left\{ c_1 \begin{bmatrix} 1 \\ 2 \\ 1 \\ 2 \end{bmatrix} + c_2 \begin{bmatrix} 1 \\ 2 \\ 1 \\ 3 \end{bmatrix} / c_1, c_2 \in R \right\}$$

(or)

$$c(A^T) = \left\{ c_1 \begin{bmatrix} 1 \\ 2 \\ 1 \\ 2 \end{bmatrix} + c_2 \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} / c_1, c_2 \in R \right\}$$

VECTOR SPACES

FOUR FUNDAMENTAL SUBSPACES OF A MATRICES



$$\text{Basis for } C(A^T) = \left\{ \begin{bmatrix} 1 \\ 2 \\ 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} \right\}; \dim C(A^T) = \text{Rank of } A$$

$$Ax = 0 \Rightarrow Ux = 0$$

$$\begin{bmatrix} 1 & 2 & 1 & 2 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ t \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$x + 2y + z + 2t = 0 \Rightarrow x = -2y - z$$

$$t = 0$$

VECTOR SPACES

FOUR FUNDAMENTAL SUBSPACES OF A MATRICES



Special solution

$$\begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix} \begin{bmatrix} -1 \\ 0 \\ 1 \\ 0 \end{bmatrix}$$

$$N(A) = \left\{ c_1 \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix} + c_2 \begin{bmatrix} -1 \\ 0 \\ 1 \\ 0 \end{bmatrix} / c_1, c_2 \in R \right\}$$

$$\text{Basis for } N(A) = \left\{ \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \\ 1 \\ 0 \end{bmatrix} \right\}; \dim N(A) = 2$$

VECTOR SPACES

FOUR FUNDAMENTAL SUBSPACES OF A MATRICES



$$A^T x = 0$$

$$A^T = \begin{bmatrix} 1 & 1 & 3 \\ 2 & 2 & 6 \\ 1 & 1 & 3 \\ 2 & 3 & 7 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 3 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 1 & 1 \end{bmatrix}$$

$$\sim \begin{bmatrix} 1 & 1 & 3 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = U^l$$

$$U^l y = 0 \Rightarrow x + y + 3z = 0 \Rightarrow x = -y - 3z$$

$$y + z = 0 \Rightarrow y = -z$$

Special solution

$$\begin{bmatrix} -2 \\ -1 \\ 1 \end{bmatrix}$$

$$N(A^T) = \left\{ c_1 \begin{bmatrix} -2 \\ -1 \\ 1 \end{bmatrix} / c_1 \in R \right\}$$

$$\text{Basis for } N(A^T) = \left\{ \begin{bmatrix} -2 \\ -1 \\ 1 \end{bmatrix} \right\} ; \dim N(A^T) = 1$$



THANK YOU

Deepthi Rao

Department of Science & Humanities